

Question	Marking Guidance	Mark	Comments
04.1	1. Bind to antigen OR Are markers; 2. (Antibodies) cause clumping/agglutination OR Attract phagocytes;	2	1. Accept opsonin for 'marker' 1. Accept form (antibody-antigen) complexes/are complementary to antigen 2. Reject clotting
04.2	Correct answer for 2 marks 110/111/111.1;; Accept for 1 mark, correct readings from graph (5.1 and 2.1)	2	
04.3	1. <u>Mean</u> (antibody concentration) increases; 2. 1 st injection protects some mice/1 mouse/2 mice OR 1 st injection causes primary (immune) response/memory cell production; 3. 2 nd /3 rd injection protects most/all mice OR 2 nd /3 rd injection causes secondary (immune) response OR 2 nd /3 rd injection uses memory cells; 4. Because antibody at/above protective level/2.1; 5. Antibody decreased (rapidly after 3 rd injection); 6. No mice protected after 180 days OR Injections/vaccine not effective in long term OR Booster required (when antibody below protective level/after 120/180 days); 7. One mouse (after first injection) has big response/already had meningitis/antigen;	4 max	2. and 3. Accept correct reference to number of unprotected mice 4. Accept converse

04.4	1. Inject vaccine (again)/meningitis antigen/ inactive antigen/dead/living bacteria/ pathogen/use a booster; 2. (Memory cells present if) <u>faster/more</u> rapid production/<u>higher</u> concentration antibody (than 1st injection) OR Immune response is quicker (than 1st injection) OR Symptoms do not develop; 3. Add enzyme attached to (second) antibody against memory cell; 4. Colour change shows memory cell present;	2	Mark as pairs, 1 and 2, 3 and 4 Accept for inject, introduce, give, use 1. Must refer to antigen or cell, 'disease' or 'meningitis' is not enough 2. Accept converse 2. Must be a comparison 4. Ignore to detect (meningitis) antibodies
TOTAL		10	